RESEARCH PACK — PROBLEM VALIDATION & MARKET EVIDENCE

CareSync — Family Health Evidence & Coordination Agent
Open Agent Hackathon 2026 | GenAI.Works

1. HACKATHON CONTEXT (FULL REQUIREMENTS)

Event: Open Agent Hackathon 2026 by GenAI.Works. Fully online, 144 hours (22–27 October 2026). Teams of 1–5. Prize pool up to $20,000.

Mandatory integrations: Zetaris + Meterless. Must use Cursor for development. Integrate NVIDIA (Nemotron) where it contributes.

Tracks: Track 01 Fragmented Intelligence · Track 02 The Agent That Can Explain Why (strongest fit) · Track 03 Reasoning Architecture · Wildcard Tinkerer.

Scoring: Impact 30 · Technical 20 · Innovation 15 · Demo 15 · Product & UX 10 · Sponsor tech 10.

Submission: 1–4 min demo video showing Zetaris + Meterless; slide deck; public GitHub + README; explicit sponsor-integration write-up. REAL/SIMULATED labels + audit trail mandatory.

Sponsor tech: Zetaris (in-situ multi-source / document-derived data), Meterless (World Model + H-MEM for longitudinal patient/family state), NVIDIA Nemotron, Cursor.

2. PROBLEM RESEARCH — FAMILY CAREGIVER BURDEN & HEALTH DATA FRAGMENTATION

In 2025, 63 million American adults provided ongoing care to adults or children with a medical condition or disability — almost one-quarter of all U.S. adults. This is a 45% increase since 2015. Of these, ~59 million care for an adult with complex needs. Average time spent caregiving: 27 hours per week; 24% provide 40+ hours per week. 30% have been caregiving for five or more years.

Source: AARP / National Alliance for Caregiving — Caregiving in the U.S. 2025 report.

Impact on caregivers:

- 64% report high emotional stress; 45% high physical strain; 1 in 5 rate their own health as fair or poor; nearly 1 in 4 have difficulty caring for themselves.
- Nearly half experience at least one negative financial impact (stopped saving, used up short-term savings, took on debt, delayed retirement).
- Caregivers spend an average of 12 hours per month solely on healthcare coordination/admin for the care recipient (AXA Health research); parents of children with special needs or long-term conditions spend 23–29 hours/month.
- 53% of caregivers have ignored or delayed their own healthcare; 39% later received a diagnosis they believe could have been caught earlier.

Sources: AARP/NAC 2025; AXA Health 2025 caregiver survey.

Health data is severely fragmented:

- 59% of individuals had multiple online medical records or patient portals in 2024 (up from 50% in 2022). Only 7% used an application that brought information from different portals together (ASTP national survey).
- Most family caregivers are responsible for communicating with providers, yet only ~11–33% access the care recipient's patient portal. Formal shared/proxy access remains rare and cumbersome.
- Relational coordination between family caregivers and medical professionals is weak (RCI ~3.10 on a scale where higher is better).

Sources: ASTP / ONC survey data; NHATS/NSOC analyses; JAMA Network Open relational coordination study.

3. WHY CURRENT SOLUTIONS DO NOT SOLVE THE PROBLEM

Patient portals (Epic, MyChart, Cerner, etc.): Designed primarily for the patient. Caregiver proxy access is inconsistent, often requires in-person identity proofing, and does not span multiple health systems. Result: caregivers juggle multiple logins or resort to shared passwords (privacy/security risk).

Apple Health / Google Health: Excellent for the individual user's own devices and records. Structural mismatch for family caregiving: single-account assumptions, limited multi-caregiver logging, weak cross-portal aggregation, platform lock-in (Apple especially), and no longitudinal "what changed and why it matters for the next specialist visit" reasoning layer.

Dedicated caregiver apps (CareZone and similar): Reviews and academic assessments find many lack evidence base, geriatric expertise, robust security, or high usability. Average ratings around 2.7/5 in one systematic look; pricing and privacy terms are often problematic. They typically store documents but do not make them queryable with evidence-backed synthesis.

Paper, email, group chats, and spreadsheets: Still the real coordination layer for most families. High cognitive load, version chaos, and easy loss of critical facts (lab trends, medication changes, prior recommendations).

None of the above maintain a living, longitudinal world-model of the patient's health trajectory + family context, query across uploaded documents and structured sources in a governed way, and produce cited visit-prep packages that a non-clinician can trust. CareSync targets exactly that gap while enforcing hard non-diagnostic boundaries.

---

4. FIT TO SPONSOR TECHNOLOGY & TRACKS

Zetaris: Enables treating extracted facts from lab PDFs, discharge summaries, med lists, and wearable CSVs as queryable sources alongside any structured tables — without forcing centralization of raw PHI into a single proprietary store. Directly supports Track 01 and the multi-source investigation required by Track 02.

Meterless World Model + H-MEM: The longitudinal patient state (conditions, key events, meds, goals) and hierarchical memory of prior conversations and documents. Essential for "what has changed in the last 90 days that the cardiologist should know?" style questions (Track 02 / 03).

NVIDIA Nemotron: Multi-step reasoning over the timeline + safety-constrained generation of explanations and suggested questions.

Safety invariant (non-negotiable): The agent must never diagnose or replace a clinician. Every clinical fact must carry a source citation or be dropped. Prominent non-medical-advice banners required.

---

5. KEY VERIFIABLE CLAIMS SUMMARY

Claim	Evidence
63 million U.S. adult caregivers in 2025 (~1 in 4 adults); +45% since 2015	AARP / NAC Caregiving in the U.S. 2025
Avg 27 hrs/week caregiving; 24% do 40+ hrs; 30% ≥5 years	AARP / NAC 2025
64% high emotional stress; 45% physical strain; ~25% difficulty caring for self	AARP / NAC 2025
12 hrs/month average on healthcare coordination admin	AXA Health 2025 research
59% of individuals have multiple patient portals/records (2024)	ASTP national survey
Only ~7% use a tool that unifies multiple portals	ASTP survey
Only 11–33% of caregivers access care recipient's portal	HINTS / NHATS-linked studies
Many caregiver apps lack evidence, security, or usability (avg ~2.7/5)	JAGS / Northwestern commentary & reviews

This research pack is intended to be attached to or referenced by the main Prove-It Agent Instruction PDF for CareSync. All statistics are drawn from publicly reported surveys and peer-reviewed or institutional analyses current as of 2024–2026. Medical-safety constraints remain absolute.